

◦ Aim:

- To estimate the percentage of iron in haematite solution containing 0.6 gm of haematite / litre by redox titration.

◦ Apparatus:

- Burette, pipette, beaker, glass rod.

◦ Chemicals:

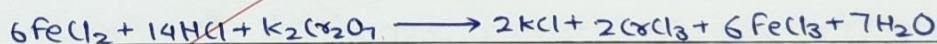
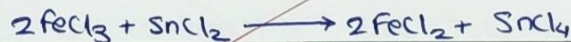
- Haematite solution, stannous chloride, mercuric chloride, potassium dichromate, diphenylamine, phosphoric acid, hydrochloric acid, sulphuric acid.

◦ Theory:

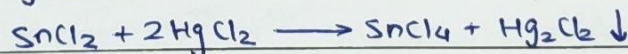
- Haematite is an oxide of iron, existing as ferric oxide (Fe_2O_3). On treating with conc. HCl, ferric oxide dissolves in the acid as ferrous oxide



- Since iron in FeCl_3 is already in the highest possible oxidation state (Fe^{3+}), it cannot be oxidized further to a stable higher oxidation state. Hence, it is first reduced to ferrous state and then titrated with potassium dichromate solution, which is an oxidation agent. Ferric chloride is reduced by stannous chloride in presence of HCl.



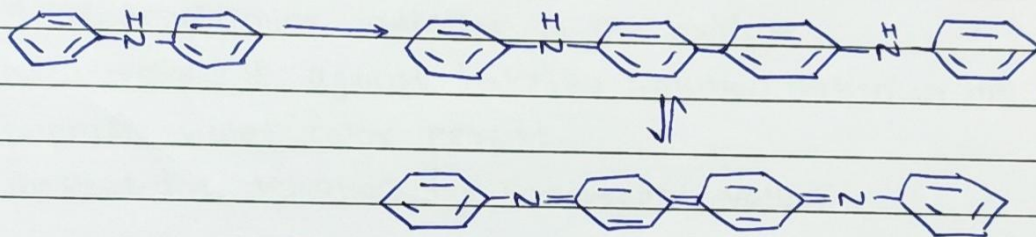
- A slight excess of SnCl_2 is added to ensure complete reduction of FeCl_3 to FeCl_2 . Since SnCl_2 can reduce $\text{K}_2\text{Cr}_2\text{O}_7$, the excess of SnCl_2 is destroyed by adding a strong solution of mercuric chloride.



- A silky white precipitate of Hg_2Cl_2 is obtained. A black precipitate of finely divided Hg is produced if large excess of SnCl_2 is present or HgCl_2 is added slowly. This finely divided Hg produced will reduce the $\text{K}_2\text{Cr}_2\text{O}_7$ and also reduces the Fe^{+3} ions to Fe^{+2} ions slowly in due course.



- The titration must be discarded if a black precipitate is obtained. The trial must also be discarded if no ppt is obtained on addition of Hg_2Cl_2 because of the complete reduction of Fe^{+3} ion to Fe^{+2} ion isn't ensured.
- As the titration involves oxidation-reduction reactions, it is called a 'redox titration'. The indicator used for detection of the endpoint is a redox indicator called diphenylamine. In presence of a strong oxidizing agent, it first undergoes irreversible chemical oxidation to give colorless diphenylbenzidine, which is the real indicator and is reversibly further oxidized to diphenylbenzidine violet.



- When Fe^{+2} ions are exhausted in the solution at the endpoint, addition of further quantity of dichromate oxidizes colorless diphenylbenzidine to diphenylbenzidine violet. Therefore, the endpoint is indicated by the color change from green (due to Cr^{+3}) to greenish violet. But the color change is sharp and detectable only when the formal potential of the indicator is intermediate of the solution titrated ($\text{Fe}^{+3}/\text{Fe}^{+2}$) and that of the titrant ($\text{Cr}^{+6}/\text{Cr}^{+3}$) and should differ by about at least 0.15V.

Observations:

Burette readings	Trial-1	Trial-2
Initial Reading	0	3.5 mL
Final Reading	3.5 mL	7 mL
Volume of $K_2Cr_2O_7$ (mL)	3.5 mL	3.5 mL

Calculations:

Volume of $K_2Cr_2O_7$ (V_1) = 3.5 mL

Normality of $K_2Cr_2O_7$ (N_1) = 0.05 N

Volume of haematite solution (V_2) = 25 mL

Normality of haematite solution:

$$N_2 = \frac{N_1 \times V_1}{V_2} = \frac{(3.5 \text{ mL})(0.05 \text{ N})}{25 \text{ mL}} = 7 \times 10^{-3} \text{ N}$$

Weight of iron per litre of solution:

$$W = N_2 \times 55.85 = (7 \times 10^{-3})(55.85) = 0.39095 \text{ gm.}$$

% of iron in haematite = $\frac{W}{0.6} \times 100 = \frac{0.39095}{0.6} \times 100 = 65.158\%$

from the other two. The formal potentials of $\text{Cr}^{+6}/\text{Cr}^{+3}$, $\text{Fe}^{+3}/\text{Fe}^{+2}$ and indicator are 1.10, 0.77, and 0.76V respectively. Therefore, the potential of $\text{Fe}^{+3}/\text{Fe}^{+2}$ system is lowered to a value of 0.6V by the addition of sulphuric acid and phosphoric acid mixture.

o Procedure:

- A known concentration of $\text{K}_2\text{Cr}_2\text{O}_7$ is filled in the burette.
- Pipette out 25ml of haematite solution into a clean conical flask.
- Add about 3ml of conc. HCl and heat it to a boil.
- To the hot mixture, add SnCl_2 solution drop-by-drop until the yellow color is discharged completely. Add 1-2 drops in excess to ensure the complete reduction of iron.
- Cool the product and add 5ml of HgCl_2 solution at a stretch, now one should see a silky white ppt.
- If a black ppt / no ppt is obtained, then discard the product.
- Add 50ml of 3M sulphuric acid, 5ml of phosphoric acid, and 3 drops of diphenylamine indicator to the mixture.
- Now, titrate it against $\text{K}_2\text{Cr}_2\text{O}_7$ solution taken in the burette till a greenish violet color persists.
- Repeat the titration for concordant values.

o Result:

- The percentage of iron in given haematite sample = 65.15%.